Cheaper Re-planning, not Less Re-planning: An Efficiency Study of SmolVLA on a Turing GPU

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Abstract

A vision-language-action (VLA) policy that re-plans at every control step pays for the whole model at every step. This report measures where that cost goes for the released LIBERO checkpoint of SmolVLA on a Tesla T4 and tests, on identical paired episodes, which reductions preserve closed-loop success. A re-plan takes 789 ms, dominated by emulated bfloat16 arithmetic and a launch-bound 10-step flow loop. Re-planning less often (open-loop chunks, gripper-event triggers, a 5-step-old image) lowers success by 12.0–12.5 points (Holm $p \leq 0.018$). Re-planning more cheaply (float16, 3 Euler steps, a CUDA-graph replay with identical actions) cuts a re-plan to 112 ms ($7.0\times$) and is non-inferior on 500 paired episodes (78.4% vs 75.4%, 95% CI of the difference $[-0.6, +6.6]$ points); one Euler step (93 ms, $8.5\times$) is non-inferior on 300 fresh pairs (80.3% vs 74.3%, CI $[+0.6, +11.4]$). Reusing one flow noise per episode removes gripper chatter but lowers success from 77.0% to 23.5%. Code: https://github.com/Ngoc-LM/efficient-vla.

1. Introduction

VLA policies map images, a language instruction and the robot state to actions through a pretrained vision-language model (VLM) and an action head [1, 17]. At batch size one inside a control loop, every re-plan waits for the full model. Efficient-VLA work reduces this cost with smaller backbones, fewer tokens or layers, cheaper action decoding, asynchronous execution or faster inference systems (Sec. 2), but a saving can change closed-loop behaviour: an ONNX export of the SmolVLA checkpoint used here cut median latency by 40% and, through truncated instructions, LIBERO-Spatial success from 70% to 41% [6].

This report studies SmolVLA [17] on LIBERO-Spatial [12] with a Tesla T4, a GPU without bfloat16 tensor cores, in a training-free setting: weights, architecture and protocol stay fixed, and only the execution changes. It surveys efficient VLAs and diffusion versus flow matching (Sec. 2; 10-paper literature matrix), reproduces the released checkpoint (Sec. 3), analyses it (Sec. 4), and tests on identical episodes whether to re-plan less often or more cheaply (Sec. 5). The levers that make re-planning cheaper are known [13, 19]; the contribution is a measured account of why each helps on this hardware and a paired closed-loop test of whether success is preserved.

Lever	Examples	Reported effect (hardware)
Smaller model	SmolVLA [17], TinyVLA [20]	0.45B, LIBERO avg 87.3; 14 vs 292 ms per action vs OpenVLA (A6000)
Fewer tokens or layers	VLA-Cache [21], EfficientVLA [22], DeeR [23]	$1.63\times$ (4090); $1.93\times$ at 28.9% FLOPs, -0.6 SR (A40); LLM FLOPs 5.2–$6.5\times$ lower (trained)
Action decoding	FAST [15], OFT [7], CogACT [10]	$\leq 5\times$ faster training, 750 vs 100 ms per chunk; $26\times$ throughput; VLM once, small DiT denoises
Execution	RTC [2], A2C2 [16]	>300 ms delay tolerated, +21 ms per chunk; 4.7 ms trained head per step
Systems	Realtime-VLA [13], ONNX [6], Coda [19]	π_0 106.5→27.3 ms (4090); -40% latency, SR drop; 2 flow steps + learned corrector

Table 1. Efficiency levers, as reported by each paper.

2. Efficient VLAs and Action Generation

Efficiency levers. Tab. 1 groups efficiency work by the part of the inference loop it changes. Token and layer reduction assumes redundancy across frames [21] or easy situations [23] and saves compute only in the language model. Chunked and asynchronous execution assumes that actions predicted from an older observation stay valid; SmolVLA's asynchronous mode finished tasks 30% faster but lowered success from 78.3% to 73.3% [17]. Systems work assumes the model stays numerically the same, which an export can violate [6]. Two gaps remain: speed-ups are reported next to aggregate success rates, though a non-significant difference is not equivalence, and most latencies come from GPUs with native bfloat16 (RTX 4090, A100, A40).

Diffusion versus flow matching. Both turn Gaussian noise $\boldsymbol{\epsilon}$ into an action chunk $\mathbf{A}$ conditioned on the observation $\mathbf{o}$. Diffusion heads [3, 10] noise the data as $\mathbf{A}_k = \sqrt{\bar{\alpha}_k}\mathbf{A} +$

$\sqrt{1-\bar{\alpha}_k}\boldsymbol{\epsilon}$ and predict the noise [4]; flow-matching heads [1, 17] interpolate linearly, $\mathbf{A}^\tau = \tau\boldsymbol{\epsilon} + (1-\tau)\mathbf{A}$, and regress the velocity of that path [11]:

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{A}_k, k \mid \mathbf{o})\right\|^2, \quad (1)$$

$$\mathcal{L}_{\text{FM}} = \mathbb{E}\left\|(\boldsymbol{\epsilon} - \mathbf{A}) - v_\theta(\mathbf{A}^\tau, \tau \mid \mathbf{o})\right\|^2. \quad (2)$$

They differ in four respects. (i) Target: noise versus velocity. (ii) Path: the linear path gives each training pair a constant velocity and yields straighter sampling trajectories [11]; on RoboTwin, $\pi_{0.5}$ keeps 71.3% success with 3 flow steps against 71.5% with 10, and drops to 59.8% with one [19]. (iii) Sampler: Diffusion Policy uses 100 DDPM steps in simulation and 10 DDIM [18] steps on the robot, CogACT 10 DDIM steps with guidance, flow models 10 Euler steps. (iv) Architecture: Diffusion Policy repeats its whole denoiser; CogACT, π_0 and SmolVLA run the VLM once and repeat only a head on its cached output.

3. Reproduction

Model and setup. The study runs the released LeRobot implementation of SmolVLA (v0.6.1) with the checkpoint `HuggingFaceVLA/smolvla_libero` on Kaggle (2× Tesla T4, PyTorch 2.11, MuJoCo 3.3.7, EGL rendering). The checkpoint differs from the paper configuration [17]: it keeps all 32 decoder layers of SmolVLM2-500M instead of 16, uses an action expert of width $0.5\times$ instead of $0.75\times$, and executes one action per predicted chunk (n_{exec}=1). It has 605M parameters (vision encoder 86M, connector 12M, language model 362M, action expert 97M) stored in bfloat16.

Pipeline. At control step k the policy receives two RGB images (third-person and wrist, resized with padding to 512^2), the instruction (at most 48 tokens) and an 8-D state (end-effector pose, two gripper joints). SigLIP encodes each image, a pixel-shuffle connector shortens it to a few tokens, and the language model processes the prefix [image; language; state] once, caching its keys and values per layer. The action expert, whose layers alternate cross-attention to this cache and self-attention over action tokens, predicts a chunk $\mathbf{A} \in \mathbb{R}^{50\times 7}$ (end-effector delta pose and gripper). It is trained with $\mathcal{L}_{\text{FM}}$ (Eq. (2)), with $\tau \sim \text{Beta}(1.5, 1)$ rescaled to $[0.001, 1]$. At inference, S=10 Euler steps $\mathbf{A} \leftarrow \mathbf{A} - \frac{1}{S} v_\theta(\mathbf{A}, \tau \mid \mathbf{o})$ integrate from τ=1 (noise) to 0; a re-plan thus costs one prefix pass and S expert passes. The first action is unnormalised and executed at 20 Hz.

Metric and protocol. Success rate (SR) is the fraction of episodes whose goal predicate (*e.g.* `On(bowl, plate)`) holds within 280 steps. An episode is a (task, stored initial state) pair; the 10 tasks run on initial states 0–19 (200 episodes; 0–49 for the main comparison). Methods are compared on identical episodes with common random numbers: the flow noise at step k is seeded by (episode, k).

Component (one re-plan)	GFLOP	bf16 (ms)	fp16 (ms)
Vision encoder, 2 cameras	431	230	39
Prefix incl. vision (32 LM layers)	527	304	82
One expert pass (flow step)	11.5	51	52
Re-plan, S=10 (released)	642	814	605
Simulator step (CPU + EGL)	–	32	

Table 2. Cost of one re-plan on a Tesla T4 (isolated components; Tab. 3, from a separate run, agrees within 3%).

Differences and result. MuJoCo is pinned below 3.4, from which the stored initial states of task 5 settle differently and SmolVLA drops from 80% to 28% on that task [9]; the evaluation uses 20 instead of 10 episodes per task. The reproduced SR of 77.0% [70.7, 82.3] (Wilson 95% CI) is below the 90% reported for the paper configuration but inside the 70–83% obtained by others with this checkpoint [6, 8].

4. Analysis

Computational bottlenecks. Tab. 2 breaks down one re-plan (CUDA events, warm-up, median of 60 calls). The simulator is not the bottleneck (32 ms per step against 814 ms for the policy). The vision encoder takes 230 ms in the released bfloat16 weights but 39 ms in float16: Turing GPUs have no bfloat16 tensor cores and emulate it. The action expert accounts for 18% of the FLOPs of a re-plan but 63% of its latency (10×51 ms), and its step time is the same in both precisions: a pass over 50 action tokens is only 11.5 GFLOP but launches several hundred kernels, so at batch one the GPU waits on launches rather than arithmetic.

Successful and failed trajectories. Successes finish in a median of 100 steps; all 46 failures run to the 280-step limit after at least one grasp. From the executed gripper commands (object poses are not logged), 28 failures release the bowl off target after one or two grasps, 9 still hold it at the limit, and 9 grasp three or more times; task 3 (bowl on the cookie box) fails in 10 of 20 episodes. Fresh flow noise at every re-plan makes consecutive chunks differ about as much as independent samples (first normalised action: 1.31 from the shifted previous chunk, 1.37 between two noises; 581 LIBERO-Object steps). The gripper command chatters accordingly: failed episodes contain 12.6 gripper segments of at most 3 steps per 100 steps, successful ones 2.7.

Strengths. **S1 Small footprint:** 1.16 GB of weights and 21 MB of activations per re-plan (float16) fit a 16 GB T4 with room for several processes. **S2 Prefix caching:** the expert reuses the cached prefix, so a flow step costs 52 ms against 304 ms for the prefix. **S3 Few-step flow:** with S=3

the first action deviates from $S{=}10$ by 0.32, a quarter of the sampling spread (1.37), and SR holds down to $S{=}1$ (Sec. 5).

Limitations. **L1 Precision mismatch:** bfloat16 weights run emulated on Turing. **L2 Launch-bound expert:** the flow loop dominates latency at batch one while using little arithmetic. **L3 Need for fresh images:** executing 5 actions per chunk, or re-planning with fresh proprioception but a 5-step-old image, loses 12.0–12.5 points (Sec. 5), so an episode needs a full re-plan at every step. **L4 A fresh sample at every step:** independent noise at every re-plan produces gripper chatter, concentrated in failures, although removing it does not raise SR (Sec. 5).

5. Improvement: Re-plan Cheaper, not Less

Bottleneck and hypotheses. A successful episode (100 steps) spends about 79 s in the policy and 3 s in the simulator. **H1 (re-plan less):** computation can be shared across steps without lowering SR, by executing k actions per chunk (open loop), by also re-planning when the next queued action switches the gripper, or by re-planning with fresh proprioception on image and language keys and values cached for 5 steps (stale image). **H2 (re-plan more cheaply):** each re-plan can be made cheaper while computing the same function up to precision and discretisation: float16 removes the emulated arithmetic (same weights, projections kept in float32), $S{=}3$ removes discretisation slack (S3 in Sec. 4), and a CUDA graph removes the launches of the expert loop, which should gain more once precision is fixed. For the graph, the prefix runs eagerly and the S-step Euler loop is captured once per S with static buffers for noise, prefix cache and time steps, then replayed; on isolated calls its actions equal the eager ones exactly in all 8 settings. A probe tests whether the chatter of L4 matters by reusing one flow noise for every re-plan of an episode (fixed noise).

Protocol. Differences in SR are paired (exact McNemar test, Newcombe 95% CI [14], Holm adjustment [5] over comparisons with the released policy); a variant is non-inferior if the lower bound exceeds -5 points, a margin fixed in advance. Latency is measured on isolated calls, because the closed-loop runs share each GPU between two processes, and checked in the loop with one process per GPU.

Re-planning less lowers success (H1 rejected). Calling the vision encoder about five times less often costs 12.0–12.5 points of SR (Tab. 4); with $k{=}10$ the drop of 9.5 points is not significant after adjustment. The event trigger removes most chatter (1.2 short gripper segments per 100 steps against 5.0) without recovering success; fresh proprioception does not replace a current image.

	bfloat16 (released)		float16	
Euler steps S	eager	graph	eager	graph
10 (released)	789	572	616	178
3	433	372	245	**112**
2	386	345	191	103
1	335	318	136	93
Fit $a{+}bS$	$284{+}50.5S$	$289{+}28.3S$	$84{+}53.3S$	$84{+}9.4S$

Table 3. Latency of one re-plan (ms, median of 60 calls, T4, batch 1); least-squares fit over S (residuals $\leq$2.7 ms).

Variant	Vis./100	SR	ΔSR [95% CI]	p_{Holm}
Released (bf16, $S{=}10$)	100	77.0	–	–
H1: re-plan less				
Open loop, $k{=}5$	20.1	64.5	-12.5 $[-20.3, -4.5]$	0.018
Open loop, $k{=}10$	10.2	67.5	-9.5 $[-17.3, -1.6]$	0.124
$k{=}5$ + gripper events	21.5	65.0	-12.0 $[-18.7, -5.2]$	0.007
Stale image ($k{=}5$, $S{=}3$)	24.6	65.0	-12.0 $[-19.1, -4.7]$	0.013
H2: re-plan more cheaply				
bf16, $S{=}3$	100	79.0	$+2.0$ $[-3.8, +7.8]$	1.000
fp16, $S{=}3$	100	77.0	0.0 $[-5.8, +5.8]$	1.000
fp16, $S{=}2$	100	77.5	$+0.5$ $[-5.7, +6.7]$	1.000
fp16, $S{=}1$	100	79.5	$+2.5$ $[-4.2, +9.2]$	1.000
Probe: fixed noise ($S{=}3$)	100	23.5	-53.5 $[-60.3, -45.4]$	$<$0.001

Table 4. Closed-loop results on 200 paired LIBERO-Spatial episodes (initial states 0–19); Holm adjustment over the nine comparisons. Vis./100: vision-encoder passes per 100 control steps.

Re-planning more cheaply preserves success (H2). Along released $\rightarrow$ float16 $\rightarrow$ $S{=}3$ $\rightarrow$ graph, a re-plan takes 789, 616, 245 and 112 ms, factors of 1.28, 2.51 and 2.19 ($7.05\times$ in total; Tab. 3). By the fits, an Euler step costs 51–53 ms eagerly in either precision (launch-bound) but 28.3 and 9.4 ms with the graph, which exposes the emulated arithmetic: the graph gains $1.38\times$ in bfloat16 and $3.46\times$ in float16 at $S{=}10$. On 200 pairs the lower bound for float16 $S{=}3$ (-5.8) misses the margin; on the planned 500 pairs (initial states 0–49) it succeeds in 78.4% against 75.4% ($+3.0$ points, 95% CI $[-0.6, +6.6]$; task-cluster bootstrap $[-1.2, +8.4]$): non-inferior, with a non-significant gain ($p{=}0.13$). Fewer steps do not lower SR measurably either ($S{=}2$: 77.5%, $S{=}1$: 79.5%), and on 300 fresh pairs (initial states 20–49, a test fixed in advance) $S{=}1$ succeeds in 80.3% against 74.3% ($+6.0$ points, 95% CI $[+0.6, +11.4]$, $p{=}0.04$; task bootstrap $[-1.0, +12.7]$) at 93 ms per re-plan ($8.5\times$): non-inferior, and better on this test but not robustly across tasks. In the loop, a float16 $S{=}3$ re-plan takes 275 ms eagerly and 114 ms with the graph, close to the isolated values. Replay keeps the first action of every episode; whole trajectories match the eager ones in 10 of 20 episodes, exactly as often as two eager runs (10 of 20): execution is not bit-reproducible, and the loop amplifies small differences.

Consistent noise is not the remedy. Fixed noise removes the chatter (0.5 against 3.8 short gripper segments per 100

steps) but lowers SR from 77.0% to 23.5% (111 against 4 discordant pairs, $p < 10^{-27}$), while $S{=}1$ removes it too (0.7) without lowering SR: chatter accompanies failures without causing them. A reading consistent with both, not tested here: fresh noise gives errors that average out over steps, fixed noise repeats one error at every step, and $S{=}1$ gives almost none, since it returns $\boldsymbol{\epsilon} - v_\theta(\boldsymbol{\epsilon}, 1 \mid \mathbf{o})$ and the minimiser of Eq. (2) at $\tau{=}1$ is $v = \boldsymbol{\epsilon} - \mathbb{E}[\mathbf{A} \mid \mathbf{o}]$.

Trade-offs and next bottleneck. The float16 gain is specific to GPUs without bfloat16 tensor cores; elsewhere only the step and graph gains would apply (not measured). Float16 changes actions by 0.010 on average (0.049 at most, magnitude 0.17) without overflow; the graph needs static shapes and one capture per S. With 32 ms of simulation, the loop runs at about 7 Hz (control rate 20 Hz). The vision-language prefix (84 ms) takes 75% of a re-plan at $S{=}3$ and 90% at $S{=}1$; further gains require a cheaper prefix, which changes the model and needs training.

6. Conclusion

For the released SmolVLA checkpoint on a T4, the cost of a re-plan is set less by FLOPs than by emulated bfloat16 and a launch-bound flow loop. Sharing computation across steps lowers success by about 12 points, whereas removing these overheads and the discretisation slack makes a re-plan $7.0\times$ cheaper and non-inferior on 500 paired episodes. One Euler step ($8.5\times$) is non-inferior on fresh episodes, with a gain not robust across tasks, whereas one noise per episode removes chatter and most successes, suggesting that chatter is a symptom. **Limitations:** one checkpoint, suite and GPU type, in simulation; the baseline (77.0%) is below the 90% of the paper configuration. **Future work:** a cheaper prefix (distillation into a smaller backbone), and the same paired closed-loop test on embedded GPUs and exported runtimes [6].

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